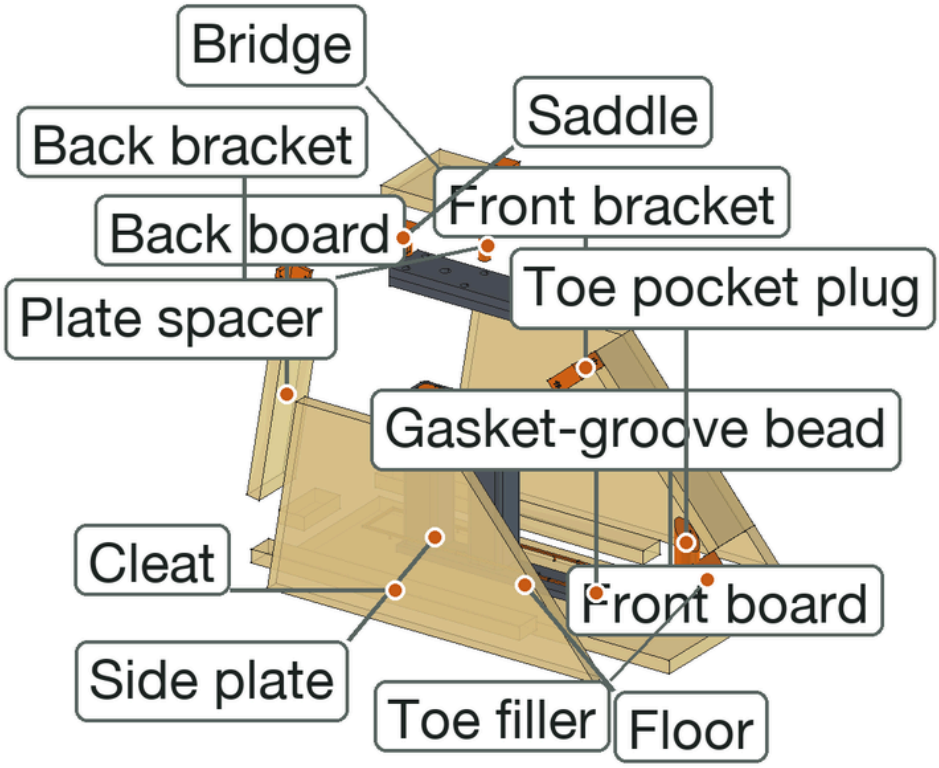
Upright molds

DIY CNC epoxy-granite mill · October 2026

How to make the upright molds. Drilling templates: `templates/upright-templates.pdf`. The sheet layout for the boards is `docs/src/boards/sheet-1.svg`; `docs/src/make_pdfs.py` also makes `molds-complete.pdf`, all the molds in one (not kept in the repo).

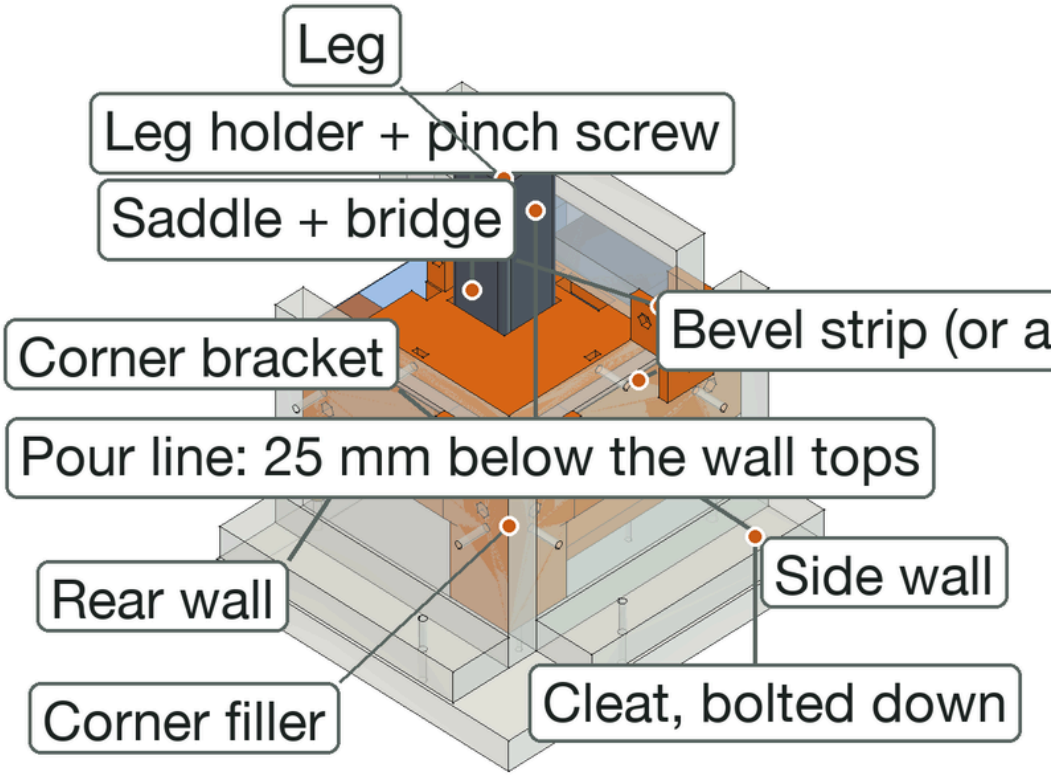


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

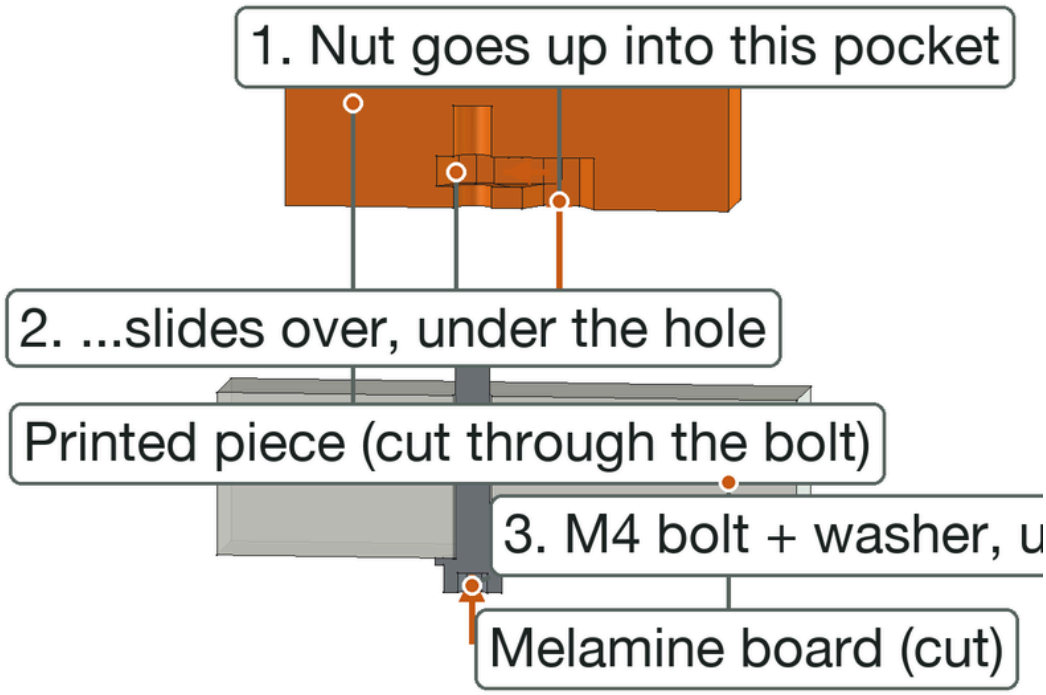
Order of work: print → cut the boards → mark them with the templates and drill → wax → assemble → pour → take apart.

1. How it works

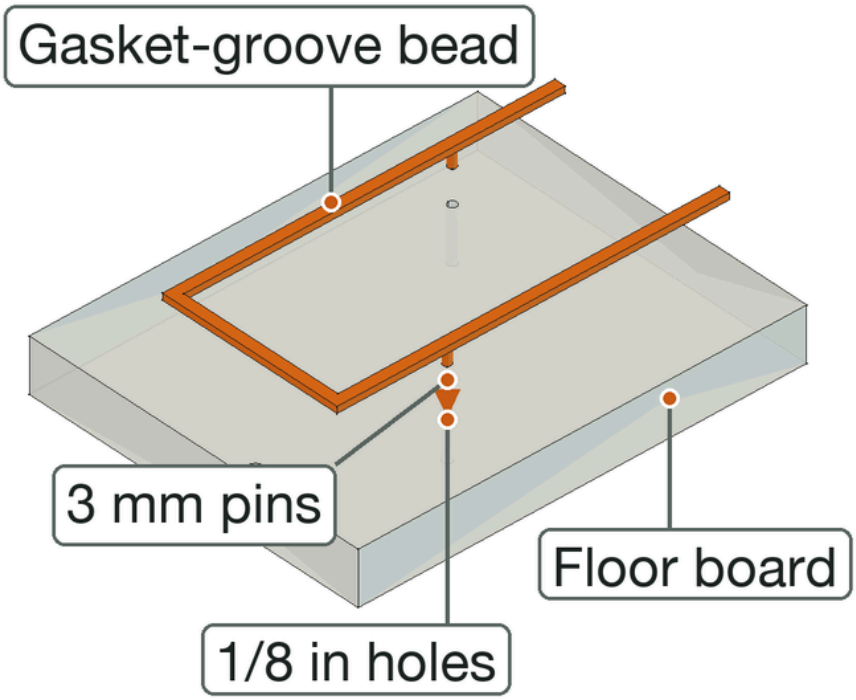
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



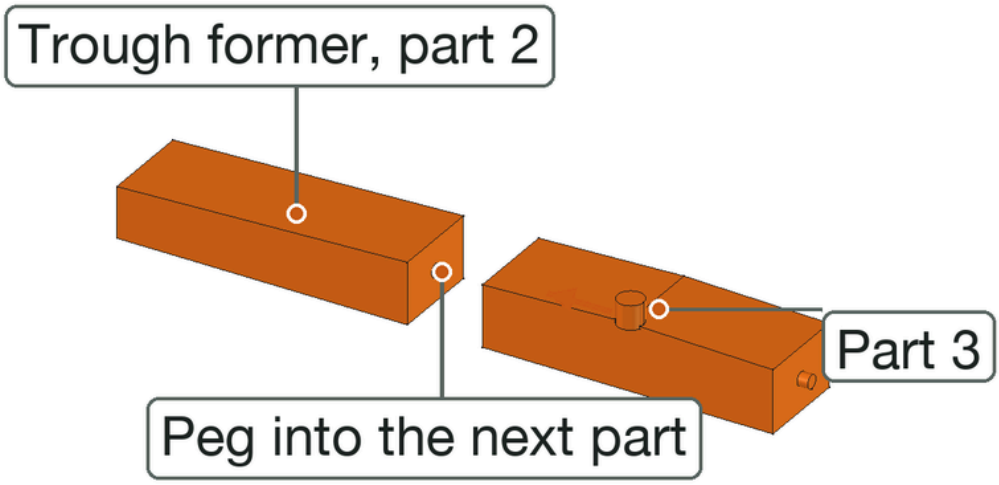
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



- **Thin pieces** (gasket-groove beads) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

Material	Amount	Notes
Melamine board, 3/4 in	about 0.4 m2	all the molds come out of 1 sheet, 4 x 8 ft
1x2 lumber (furring strip is fine)	1 x 8 ft 1x2 (0.9 m)	the cleats: they lie on the floor outside the walls
PETG	0.25 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	a few meters	the rounded outside edges
2x4 offcut	4 blocks per upright mold, about 50 mm	feet

Hardware	Qty
M4 nut	20
M4 washer	20
M4 x 30 bolt	11
M4 x 35 bolt	2
M4 x 45 bolt	7
1/2-13 x 1-1/2 in bolt + nut, greased (forms a dowel hole)	2
3/8-16 x 3 in bolt, greased (holds the top plate, both layers)	2
the two column tie rods themselves, greased: up through the floor and the columns into the top plate, to center the columns while the casting goes off	2

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a 2-1/2 in hole saw, a jigsaw for the side plates' slope, a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl.

3. Print

32 prints, 0.25 kg, about 18 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.

File	Qty	g	Time each
`upright/kit-back-bracket.stl`	4	21	27m 31s
`upright/kit-front-bracket.stl`	4	32	38m 26s
`upright/kit-saddle-flat.stl`	4	18	22m 40s
`upright/filler-toe-filler.stl`	1	26	1h 30m 59s
`upright/former-foot-gasket-groove-part1of4.stl`	1	3	17m 31s
`upright/former-foot-gasket-groove-part2of4.stl`	1	1	8m 54s
`upright/former-foot-gasket-groove-part3of4.stl`	1	1	8m 54s
`upright/former-foot-gasket-groove-part4of4.stl`	1	3	17m 27s

File	Qty	g	Time each
`upright/plug-toe-plug.stl`	1	13	1h 3m 42s
`upright/kit-plate-spacer.stl`	2	8	20m 9s
`tools/kit-pour-gauge.stl`	2	9	24m 33s
`tools/kit-screed-shoe.stl`	2	20	44m 42s
`upright/cast-in-column-ring-l1a.stl`	8	93	44m 23s

4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches). The cleats are cut from 1x2 lumber, not the sheet; the bridges and struts come out of the sheet's offcuts.

Board	Qty	Size (mm)	Notes
Upright floor	1	198 x 581	
Upright side plate, outer	1	501 x 271 (profile)	cut to the profile
Upright side plate, inner	1	501 x 271 (profile)	cut to the profile
Upright back board	1	80 x 272	ends cut square
Upright front board	1	80 x 292	ends cut square
Upright bridge	1	131 x 138	
Upright cleat, outer	1	411 x 38	1x2 lumber; flat on the floor
Upright cleat, inner	1	411 x 38	1x2 lumber; flat on the floor
Upright cleat, back	1	80 x 38	1x2 lumber; flat on the floor

5. Mark and drill

1. Print the drilling templates (`templates/upright-templates.pdf`) at **actual size** and check the 100 mm square.
2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove
4.5 mm	11/64 in	cleat, corner bracket, saddle, toe-filler, toe-plug
10 mm	25/64 in	3/8-16 bolt into the top plate
13.1 mm	33/64 in	1/2-13 bolt, base dowel hole
13.5 mm	17/32 in	column tie rod, from below

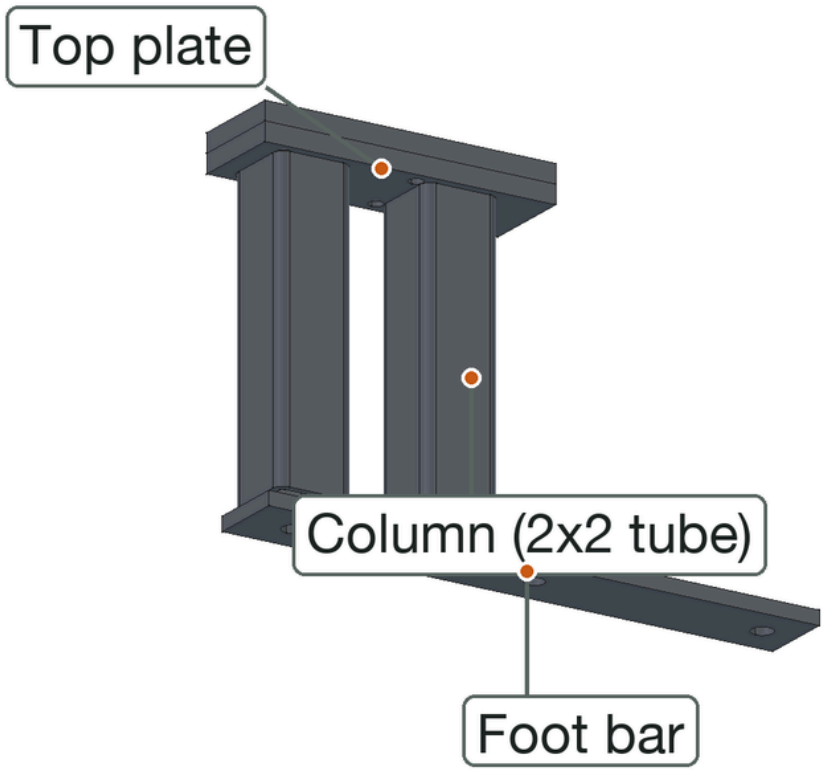
6. Wax

- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

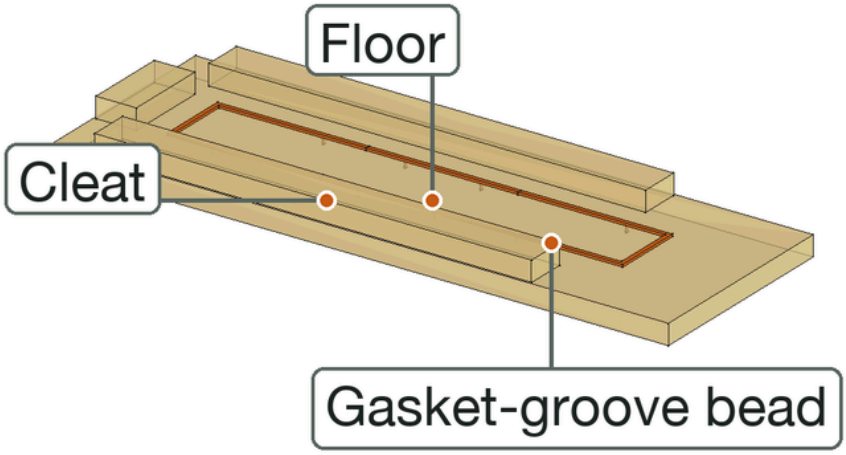
7. Assemble

Uprights (one mold, cast twice)

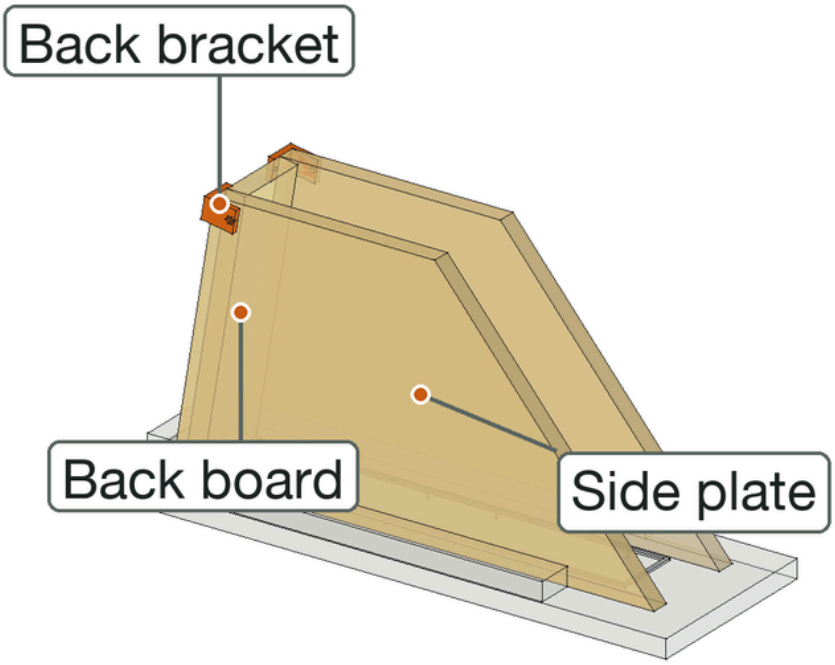
Cast foot down. The two uprights are the same part: cast one, take the mold apart after the cure, scrape and re-wax it and cast the other. The skeleton: the foot bar, two columns and the top plate (two flat bars), clamped later by the column tie rods.



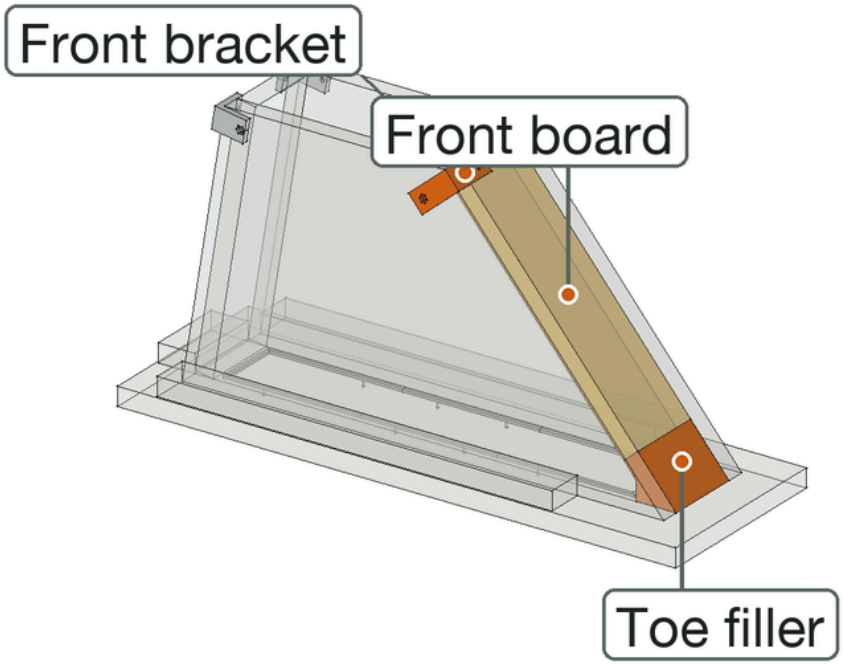
1. Gasket bead pinned on; cleats bolted down.



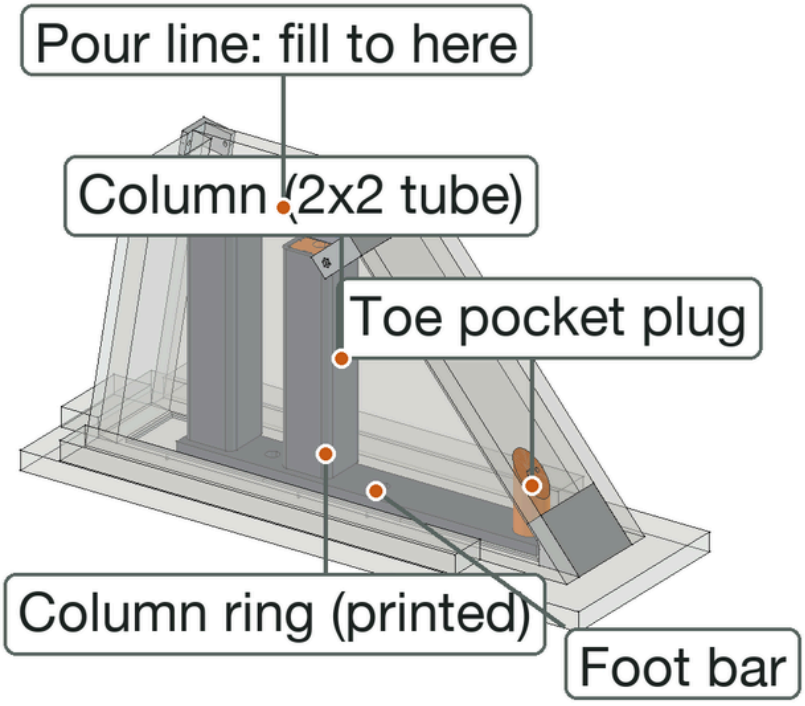
2. Side plates against their cleats, back board between them (it leans in, like the front board), two back brackets.



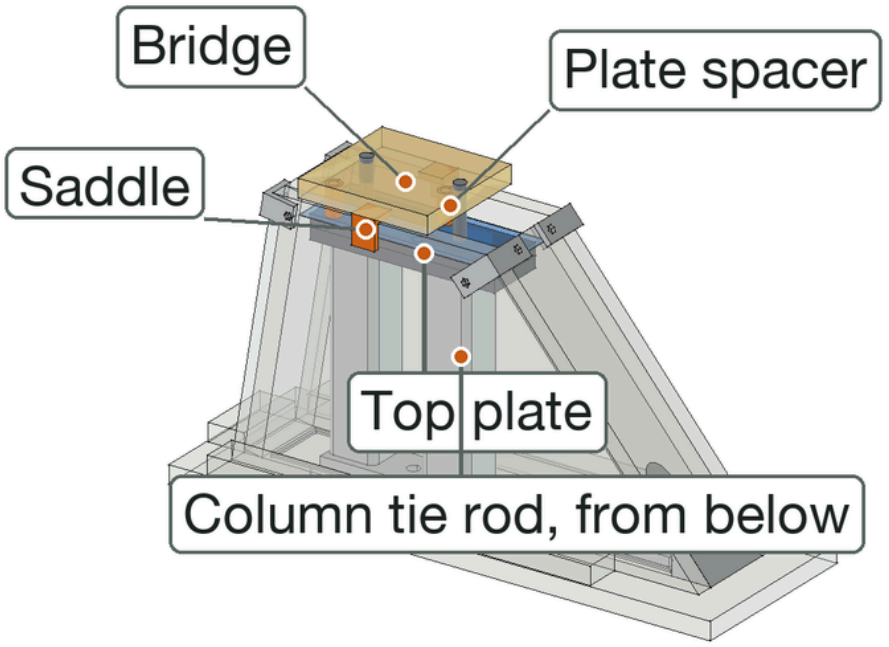
3. Toe filler bolted down, the front board's foot in its lip, two front brackets at the top.

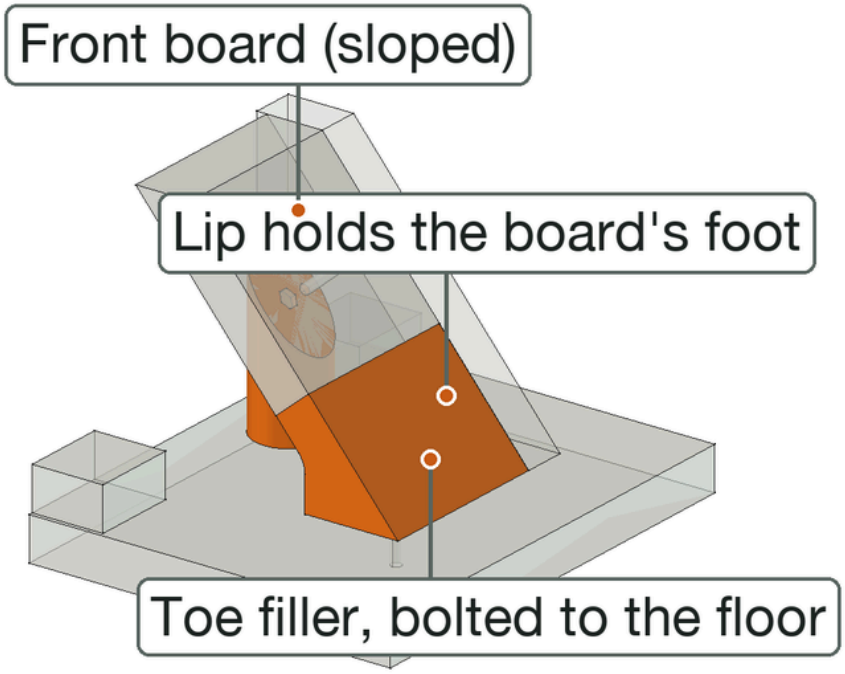


4. Mold on its four 2x4 blocks. The foot bar on the floor; the two columns on beds of thickened epoxy on it, the top plate and bridge set on them while it goes off, then lifted away; the toe pocket plug on the foot bar; the two dowel-hole bolts from underneath.



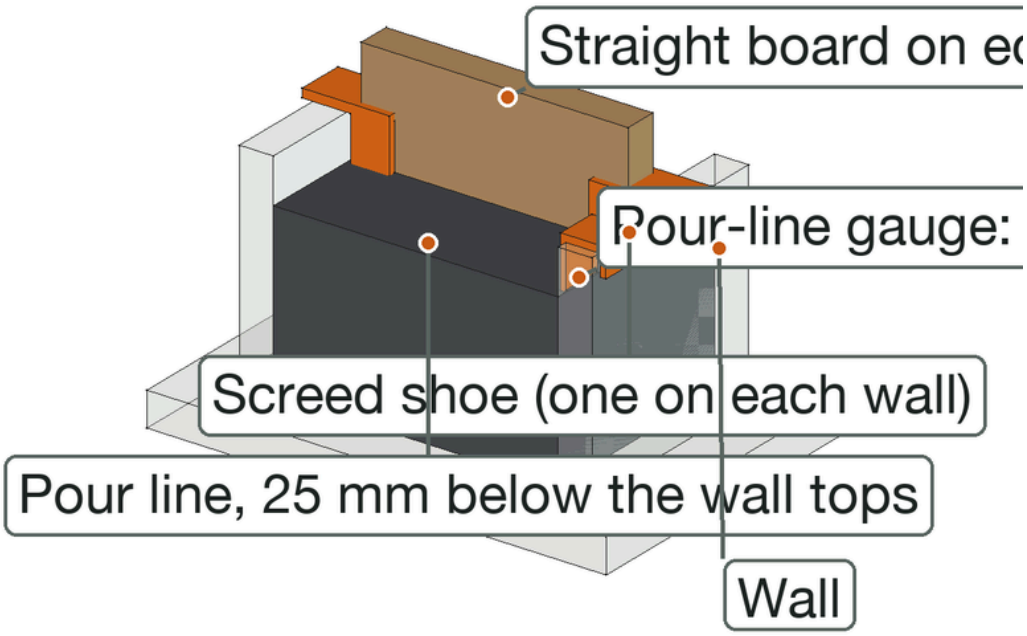
5. Fill to the line. Top plate (both bars) back on, hung from the bridge; bridge onto its saddles. Then the two column tie rods, greased, up from under the floor through the foot bar and the columns into the top plate: they center the columns while it cures.





8. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.
- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



9. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers.
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

10. If something doesn't fit

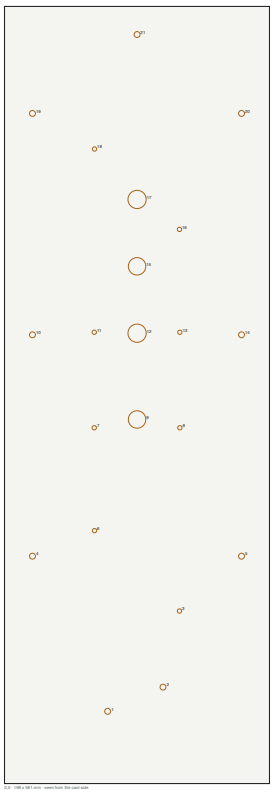
Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it

Problem	Fix
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

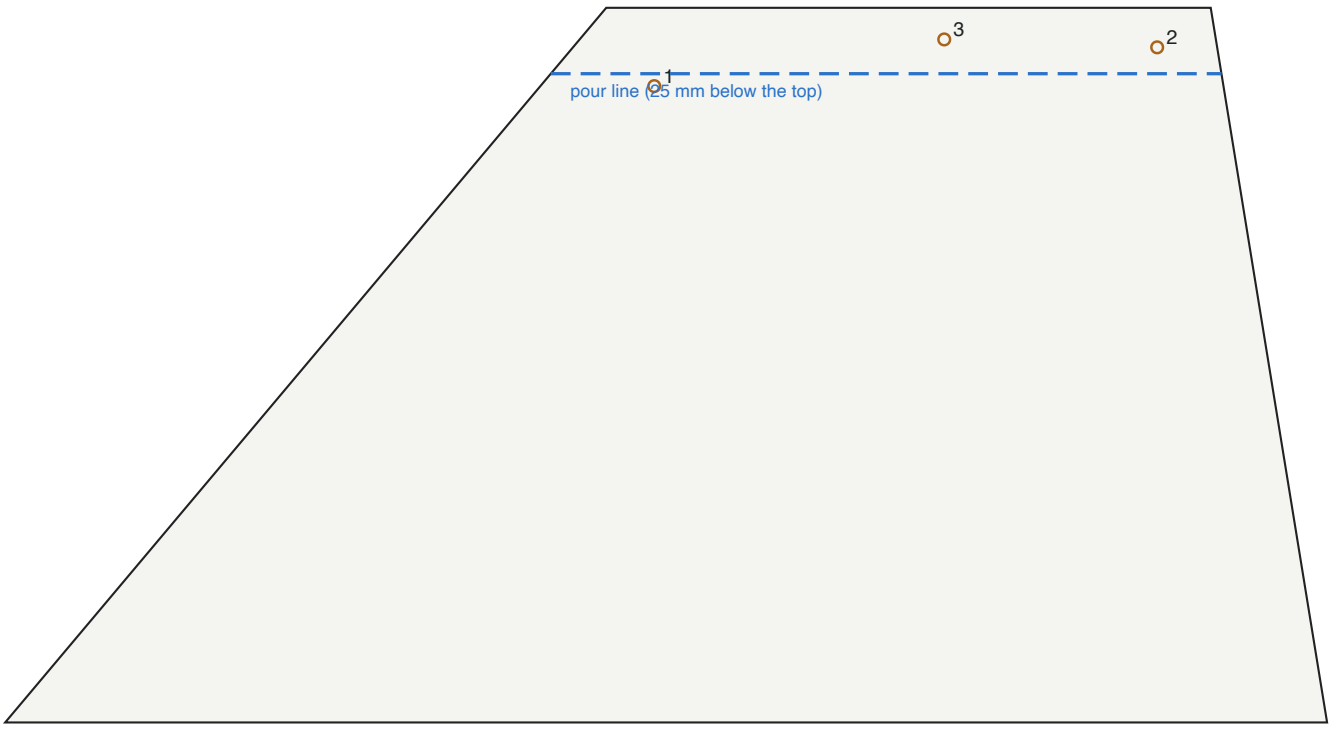
Upright floor



#	x	y	Hole	For
1	77	54	4.5	toe-filler
2	119	72	4.5	toe-filler
3	131	129	3.2	foot-gasket-groove
4	21	170	4.5	cleat
5	177	170	4.5	cleat
6	67	189	3.2	foot-gasket-groove
7	67	266	3.2	foot-gasket-groove

#	x	y	Hole	For
8	131	266	3.2	foot-gasket-groove
9	99	272	13.1	1/2-13 bolt, base dowel hole
10	21	335	4.5	cleat
11	67	337	3.2	foot-gasket-groove
12	99	337	13.5	column tie rod, from below
13	131	337	3.2	foot-gasket-groove
14	177	335	4.5	cleat
15	99	387	13.1	1/2-13 bolt, base dowel hole
16	131	414	3.2	foot-gasket-groove
17	99	437	13.5	column tie rod, from below
18	67	474	3.2	foot-gasket-groove
19	21	501	4.5	cleat
20	177	501	4.5	cleat
21	99	560	4.5	cleat

Upright side plate, outer



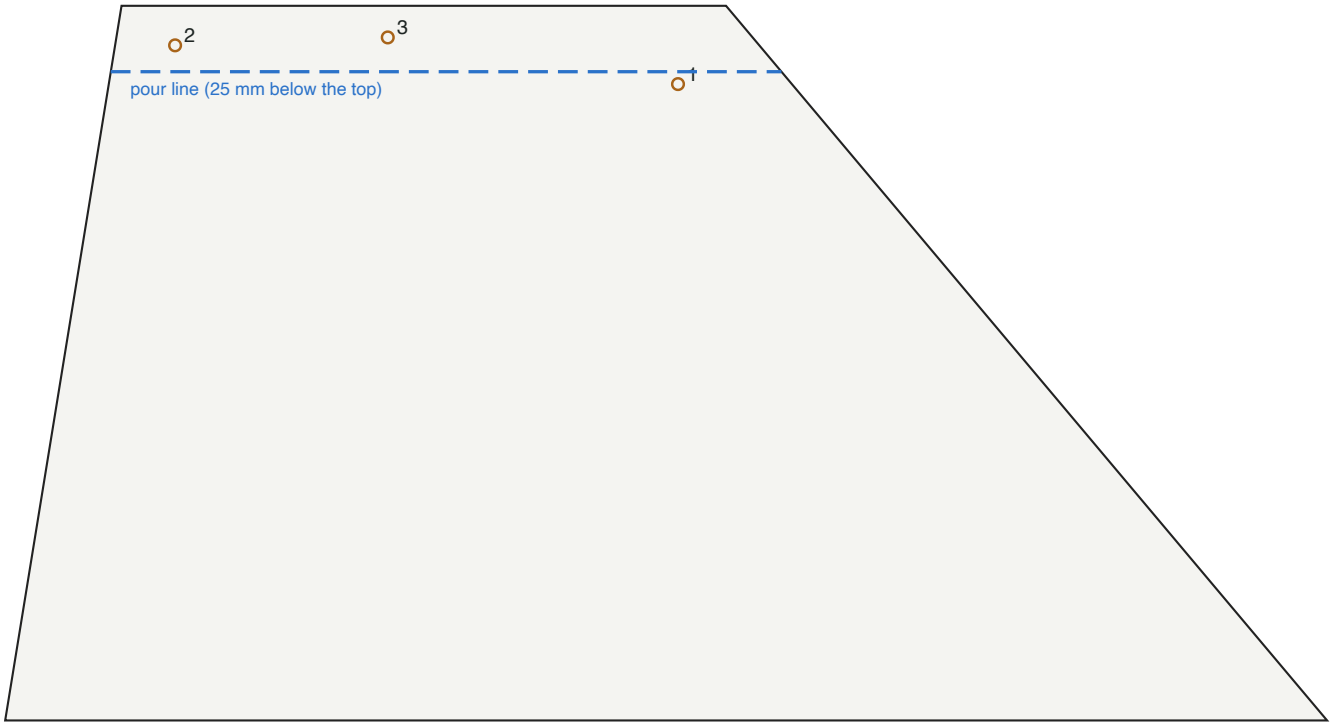
0,0 · 501 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	246	241	4.5	corner bracket
2	437	256	4.5	corner bracket

#	x	y	Hole	For
3	356	259	4.5	saddle

Outline corners (x, y): (0, 0), (501, 0), (457, 271), (228, 271)

Upright side plate, inner

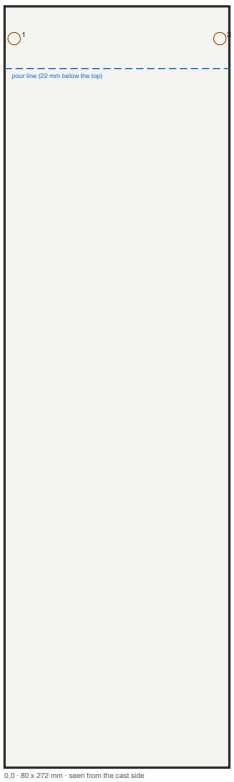


0,0 · 501 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	255	241	4.5	corner bracket
2	64	256	4.5	corner bracket
3	145	259	4.5	saddle

Outline corners (x, y): (501, 0), (0, 0), (44, 271), (273, 271)

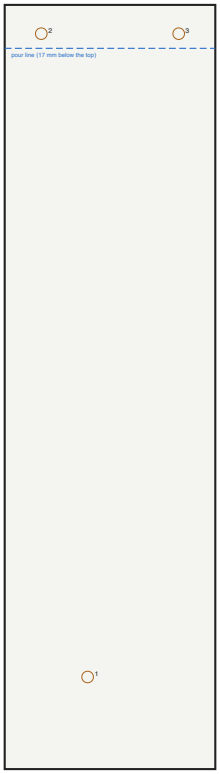
Upright back board



0,0 · 80 x 272 mm · seen from the cast side

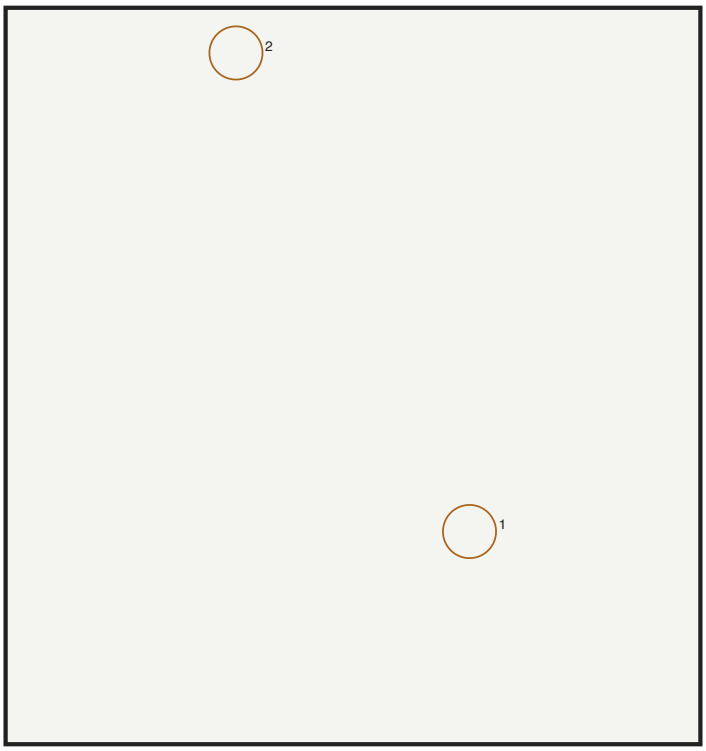
#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

Upright front board



#	x	y	Hole	For
1	32	35	4.5	toe-plug
2	14	281	4.5	corner bracket
3	66	281	4.5	corner bracket

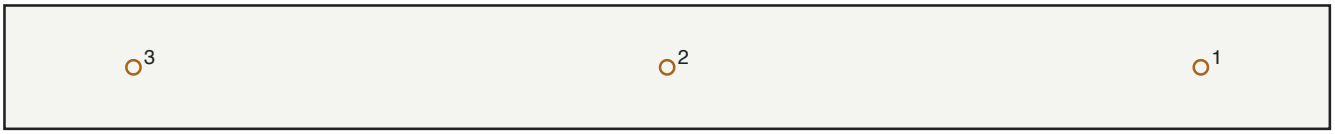
Upright bridge



0,0 · 131 x 138 mm · seen from the outside (top)

#	x	y	Hole	For
1	87	40	10	3/8-16 bolt into the top plate
2	43	130	10	3/8-16 bolt into the top plate

Upright cleat, outer



0,0 · 411 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	371	19	4.5	cleat
2	205	19	4.5	cleat
3	40	19	4.5	cleat

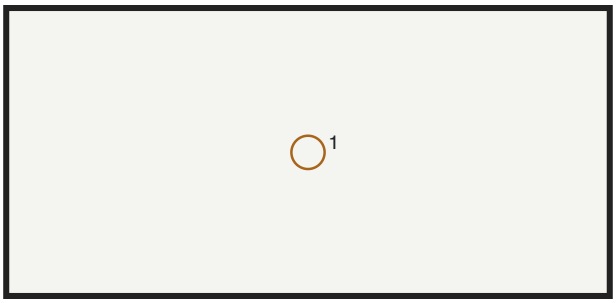
Upright cleat, inner



0,0 · 411 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	371	19	4.5	cleat
2	205	19	4.5	cleat
3	40	19	4.5	cleat

Upright cleat, back



0,0 · 80 x 38 mm · seen from the outside (top)

#	x	y	Hole	For
1	40	19	4.5	cleat